

accuracy of the regression coefficients

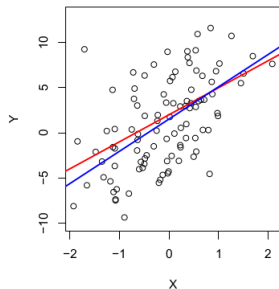
simple linear regression coefficients $\hat{\beta}_0, \hat{\beta}_1$ are determined on *TR* data **actually** generated by

$$y = \beta_0 + \beta_1 x + \epsilon$$

where ϵ is an error term considering:

- the probable not linear nature of the relationship between y and x
- there may be other variables influencing y
- there may be measurement errors
- ...

$\Rightarrow$ regression coefficients $\hat{\beta}_0, \hat{\beta}_1$ differ from **real** coefficients β_0, β_1



example:

red line is given by

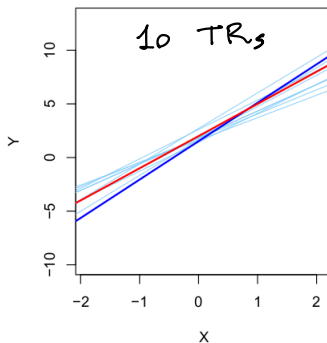
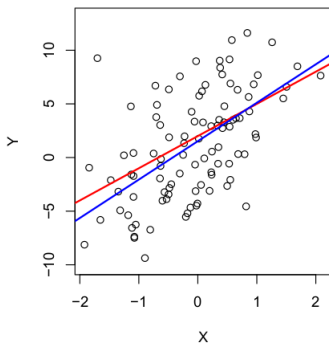
$$y = 2 + 3x$$

blue line is the regression line of observed data generated by a process of the type

$$y = 2 + 3x + \epsilon$$

regression line is an **unbiased estimator**:

β_0, β_1 **tend** to be not overestimated/underestimated by $\hat{\beta}_0, \hat{\beta}_1$



how well regression coefficients $\hat{\beta}_0, \hat{\beta}_1$ approximate unknown real β_0, β_1 ?

the average of $\hat{\beta}_0, \hat{\beta}_1$ determined over many datasets would be accurate but we have only one $TR...$

assuming all ϵ_i (associated to x^i) having common standard deviation σ_ϵ and uncorrelated, we can estimate the variance of $\hat{\beta}_0, \hat{\beta}_1$:

it can be shown that

$$\sigma_{\hat{\beta}_0}^2 = \sigma_\epsilon^2 \left[\frac{1}{m} + \frac{\bar{x}^2}{\sum_{i=1}^m (x^i - \bar{x})^2} \right] \quad \text{and} \quad \sigma_{\hat{\beta}_1}^2 = \frac{\sigma_\epsilon^2}{\sum_{i=1}^m (x^i - \bar{x})^2}$$

Handwritten notes: An arrow points from the term $\frac{1}{m}$ in the first equation to $\frac{1}{m} \sum_{i=1}^m x^i$. To the right, a graph shows a red starburst at the origin with the label "SPREAD = 0".

- $\sigma_{\hat{\beta}_1}^2$ tends to be smaller for larger spread of x^i
 - $\sigma_{\hat{\beta}_0}^2$ would be the same as the one for the sample mean if $\bar{x} = 0$
 - if σ_ϵ is unknown it can be estimated as $\sqrt{RSS/(m-2)}$ (residual standard error)
 - $\sigma_{\hat{\beta}_0}^2, \sigma_{\hat{\beta}_1}^2$ can be used to construct confidence intervals, for example:
95% confidence intervals for $\hat{\beta}_0$ and $\hat{\beta}_1$ are
- Handwritten notes:* "THE LARGER DATA SPREAD THE MORE INFO I HAVE TO ESTIMATE SLG" and "IS A SPREAD MEASURE" with an arrow pointing to the spread of x^i in the first bullet point.

$$\left[\hat{\beta}_0 - 2\sigma_{\hat{\beta}_0}, \hat{\beta}_0 + 2\sigma_{\hat{\beta}_0} \right] \quad \text{and} \quad \left[\hat{\beta}_1 - 2\sigma_{\hat{\beta}_1}, \hat{\beta}_1 + 2\sigma_{\hat{\beta}_1} \right]$$

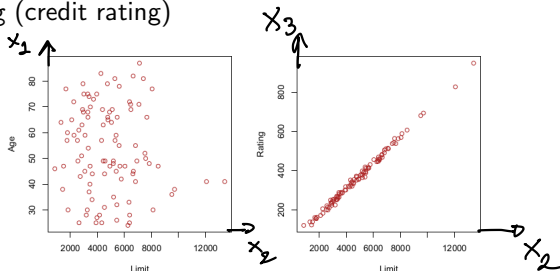


collinearity and multicollinearity

uniqueness of regression coefficients is important for interpretation... *MULTIVARIATE LINEAR REGRESSION*

collinearity refers to the situation in which two or more predictor variables are closely related to one another

example: given a dataset of credit card holders, determine variable balance y (credit card debt for each individual) based on predictors age, limit (credit limit), and rating x_1 , x_2 , x_3 (credit rating)

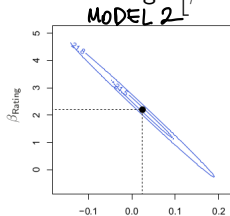
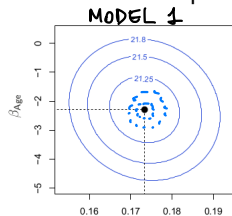


- age and limit exhibit no correlation
- rating and limit are highly correlated $\implies$ they are **collinear**!

since collinear predictors vary together it is difficult to determine their real single influence on the target variable...

consider the balance regression on age and limit (Model 1), and the balance regression on rating and limit (Model 2)

RSS level curves plotted for the range $[\hat{\beta} - 4\sigma_{\beta}, \hat{\beta} + 4\sigma_{\beta}]$ for every predictor:



MODEL 1 **IDEAL SOLUTION**

	Coefficient	Std. error
Model 1 Intercept	-173.411	43.828
age	-2.292	0.672
limit	0.173	0.005
Model 2 Intercept	-377.537	45.254
rating	2.202	0.952
limit	0.025	0.064

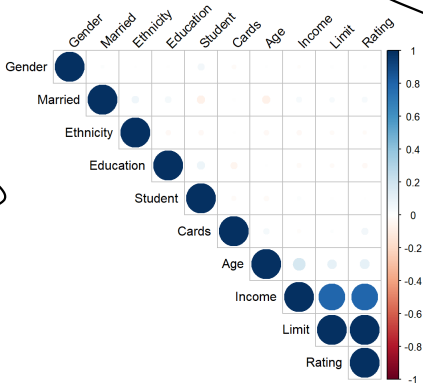
- the collinear rating-limit has level curves characterized by "narrow valleys"
- the range of limit in the collinear case is much larger
- small data change can move the regression coefficients anywhere along the narrow valley

⇒ much more uncertainty of the regression coefficients in the collinear case!



collinearity can be detected by the **correlation matrix** inspection

ELEMENT $ij =$
CORR. COEFF.
BETWEEN
FEATURE i AND
FEATURE j



SYMMETRIC

→ THE
RANGE OF
CORR. COEFF.
IS $[-1, 1]$

if two predictors are highly correlated (above a certain threshold) one can be removed from the regression model without compromising accuracy of the regression fit

multicollinearity is the collinearity between three or more predictors even if no pair of predictors has a particularly high correlation

⇒ multicollinearity cannot be captured by the correlation matrix

multicollinearity is detected by the **variance inflation factor** (VIF):

$$VIF(\hat{\beta}_j) = \frac{1}{1 - R^2_{x_j|x_{-j}}} \rightarrow R^2 \text{ ASSOCIATED TO THE LINEAR REGRESSION OBTAINED IGNORING } x_j$$

where $R^2_{x_j|x_{-j}}$ is the R^2 obtained by the regression of predictor x_j on all other predictors (x_{-j})

- VIF is always larger (or equal) than 1
- in practice VIF exceeding 5 or 10 indicate a collinearity problem

$$\text{IF } R^2_{x_j|x_{-j}} \approx 1$$

⇒ x_j IS REDUNDANT



MULTICOLLINEARITY.

in the balance regression $VIF(\text{age})=1.01$, $VIF(\text{rating})=160.67$, and $VIF(\text{limit})=160.59$

$$\text{IF } R^2_{x_j|x_{-j}} \approx 0$$

⇒ x_j PROVIDE SIGNIFICANT INFO.

⇒ NO MULTICOLLINEARITY



polynomial regression

sometimes linear regression is not suited...

...but linearity is desirable (simple training problem)

STRONGLY CONNECTED
WITH CONVEXITY OF TRAINING PROBLEM

- idea: maintain the linearity of the model with respect to the training variables β and NONLINEARLY map the features in a larger space (of dimension q)

example of $q = 3$ polynomial transformation $x \Rightarrow \begin{pmatrix} x \\ x^2 \\ x^3 \end{pmatrix}$

$\text{POLYNOMIAL } q=3 \rightarrow x^0 \quad x^1 \quad x^2 \quad x^3$

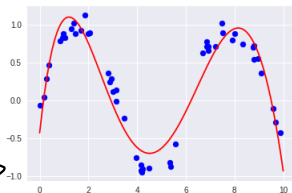
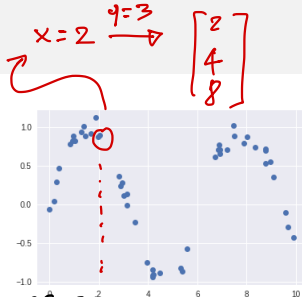
↓
LARGER
THAN n

$$y = \beta_0 + \beta_1 x \Rightarrow y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

features are nonlinearly transformed (in a 3D space) but the model remains linear in parameters β !

exercise 4.1

NUMBERS



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